SECTION 1: 220V AC MAINS & DC POWER DISTRIBUTION

Switching PSU (12V 5A), LM2596 Step-Down (5V/3A) & Dual Rails

J_AC
L
N
PE
EARTH
IEC C14
220VAC PH
F1
5A/250V
SW1
KCD1 (Mains)
MOV
10D
PSU1
AC (L)
AC (N)
FG (Earth)
+12V
GND
LRS-75-12 (5A / 60W)
220VAC IN → 12VDC OUT
F2 +12V (BUS)
5A (Blade)
+3.3V (VDD)
C1
100uF/35V
C3
100uF/10V
U2 (BUCK)
IN+ (12V) OUT+ (5V)
IN- (GND) OUT- (GND)
+5V (VIN)
C2
220uF/16V
LM2596-5.0 DC-DC
High-Efficiency 3A Step-Down
R_RST
10k
SW_RST
Reset Button

SECTION 2: MULTI-CHANNEL 12V RELAY ACTUATOR CONTROLLERS

Optocoupled 10A Relays, 1N4007 Flyback Suppression & 12V Actuator Loads

RLY_MOD1 (8-CHANNEL OPTOCOUPLED RELAY BOARD)
+5V (RELAY VCC)
GND

K1: EXHAUST FAN
12V DC 120mm Ball Bearing Fan (>70 CFM)
Flyback Diode: D5 (1N4007)
POWER: COM ← +12V
LOAD: NO → PORT 1
IN1 ← IO9

K2: INTAKE FAN
12V DC Fresh Air Intake / CO2 Purge Fan
Flyback Diode: D6 (1N4007)
POWER: COM ← +12V
LOAD: NO → PORT 2
IN2 ← IO10

K3: MISTER PUMP
12V DC Diaphragm Mist Pump (60W Peak)
Flyback Diode: D7 (1N4007)
POWER: COM ← +12V
LOAD: NO → PORT 3
IN3 ← IO11

K4: GROW LIGHTS
12V LED Grow Light Strip (Full Spectrum)
Flyback Diode: D8 (1N4007)
POWER: COM ← +12V
LOAD: NO → PORT 4
IN4 ← IO12

K5: SOLENOID VALVE
12V DC Water Solenoid (NC, 1/2" Thread)
Flyback Diode: D9 (1N4007)
POWER: COM ← +12V
LOAD: NO → PORT 5
IN5 ← IO13

Songle SRD-05VDC-SL-C (10A 250VAC / 10A 30VDC Rating)
Optically Isolated Active-LOW Coil Triggers with 1N4007 Flyback Suppression

SECTION 3: LIQUID HANDLING SENSORS & HARDWARE CONTROLS

YF-S201 Hall Flow Meter, XKC-Y25 Capacitive Level & Panel Switches

SEN_FLOW (YF-S201)
Red: VCC (5V)
Black: GND (VSS)
Yellow: Pulse Out → IO14
R9
10k
Hall-Effect Flow Sensor
1-30 L/min, Pulse: 7.5*Q

SEN_LEVEL (XKC-Y25)
Brown: VCC (5-12V)
Blue: GND (VSS)
Yellow: Digital Out → IO21
Capacitive Non-Contact
Penetrates 20mm Tank Wall

FRONT PANEL HARDWARE CONTROLS & OVERRIDES
SW_MODE (3-POS)
Rotary Override
POS 1: AUTO (Local PID)
POS 2: SYSTEM OFF
POS 3: MANUAL OVERRIDE
3-Position Heavy Duty Mode Selector
Allows Growers to Force Autonomous PID, Full Shutdown, or App Bypass

U1
ESP32-S3-WROOM-1
DevKit N16R8 (Xtensa 240MHz)
3V3 (1)
VIN (2)
GND_1 (3)
EN/RST (4)
IO9 (5)
IO10 (6)
IO11 (7)
IO12 (8)
IO13 (9)
IO14 (10)
IO21 (11)
IO38 (12)
GND_2 (13)
IO4 (14)
IO5 (15)
IO6 (16)
IO7 (17)
IO8 (18)
IO1/SDA (19)
IO2/SCL (20)
IO47 (21)
IO48 (22)
GND_3 (23)

SECTION 4: CORE PROTOTYPE ENVIRONMENTAL SENSORS

Breadboard Sensing Array: DHT11 Temp/Hum, LDR Lux, MQ-135 Gas & Submersible Water Probe

S1 (DHT11 / SHT31)
1: VCC (3.3V)
2: DATA (IO4)
3: NC (Unused)
4: GND (VSS)
R1
10k
Temp & Relative Humidity
±2% RH / 0-100% Humidity

S2 (LDR SENSOR)
1: VCC (3.3V)
2: DIVIDER JCT
3: GND (via 10k)
R2
10k
GL5528 Photocell
0-1000 Lux Light Tracker

S3 (MQ-135 MODULE)
+5V (HEATER)
1: VCC (5V VIN)
2: GND (VSS)
3: DO (Threshold)
4: AO (Analog Out)
CO2 / NH3 / Air Index
Dedicated 5V Heater Rail

S4 (WATER LEVEL)
1: + (PWR STROBE)
2: - (GND VSS)
3: S (SIGNAL OUT)
Submersible Probe
Anti-Corrosion Strobe

SECTION 5: SUSPENDED I2C ENVIRONMENTAL PROBE (WEATHER SHIELD)

True NDIR CO2 (SCD41), Precision SHT31, BH1750 Lux & SSD1306 Cabinet OLED Display

R7
4k7
R8
4k7
U3 (SCD41)
Sensirion True NDIR CO2
400-5000 ppm (±40ppm)
1: VDD (3.3V)
2: GND (VSS)
3: SDA (Data)
4: SCL (Clock)
U4 (SHT31-D)
Sensirion Humidity/Temp
±0.3°C / ±2% RH
U5 (BH1750)
Digital Lux Sensor
1-65535 Lux 16-Bit ADC
U6 (SSD1306)
0.96" OLED Display
128x64 Enclosure Screen

SECTION 6: PROTOTYPE LED FEEDBACK & EXTERNAL GX16 PORTS

On-board Diagnostics LEDs + IP65 Screw-Lock Aviation Ports for Field Equipment

D1 FANS LED
D2 VALVE/CO2
D3 MISTERS
D4 LIGHTS
LED_WF WiFi (Blue)
LED_AL Alarm (Red)
R3 220R
R4 220R
R5 220R
R6 220R
R_WF 220R
R_AL 220R

PORT 1: [EXHAUST FAN]
Pin 1: +12V (Relay 1 NO) | Pin 2: GND | Pin 3-4: Aux
PORT 2: [INTAKE FAN]
Pin 1: +12V (Relay 2 NO) | Pin 2: GND | Pin 3-4: Aux
PORT 3: [MISTER PUMP]
Pin 1: +12V (Relay 3 NO) | Pin 2: GND | Pin 3-4: Aux
PORT 4: [GROW LIGHTS]
Pin 1: +12V (Relay 4 NO) | Pin 2: GND | Pin 3-4: Aux
PORT 5: [WATER VALVE]
Pin 1: +12V (Relay 5 NO) | Pin 2: GND | Pin 3-4: Flow

SECTION 7: COMPLETE PINOUT DIRECTORY & SYSTEM NETLIST

Official Engineering Documentation, Net Connections & Protection Standards

KABUTECH HIYAS — SMART MUSHROOM AUTOMATION SYSTEM

MASTER ELECTRICAL WIRING SCHEMATIC & COMPONENT NETLIST (PROTEUS ISIS SPEC)

DOC ID: KBT-HIYAS-SCH-002
DESIGN: Zandro2005 / Kabutech R&D
MICROCONTROLLER: ESP32-S3 (Xtensa Dual-Core 240MHz)
POWER SPECS: 220VAC In / 12VDC 5A Main / 5VDC / 3.3VDC
REVISION: REV 2.4 (MASTER)
STATUS: BENCH TESTED & VERIFIED

ESP32-S3 COMPLETE PIN ASSIGNMENT REFERENCE

- GPIO 4 → DHT11 / SHT31 Data (+10kΩ Pull-up to 3.3V)
- GPIO 5 → LDR Light Sensor (10kΩ Divider Junction to GND)
- GPIO 6 → MQ-135 Analog Output AO (ADC1 CH5)
- GPIO 7 → Water Level Sensor Signal S (ADC1 CH6)
- GPIO 8 → Water Level Power V+ (Anti-Electrolysis Gated Strobe)
- GPIO 1 → Shared I2C SDA (SCD41, SHT31, BH1750, OLED)
- GPIO 2 → Shared I2C SCL (SCD41, SHT31, BH1750, OLED)
- GPIO 9 → Relay 1 / D1 LED: Exhaust Fan (12V DC)
- GPIO 10 → Relay 2 / D2 LED: Intake Fan / CO2 Valve
- GPIO 11 → Relay 3 / D3 LED: Misting Pump (60W 12V)
- GPIO 12 → Relay 4 / D4 LED: Grow Light Strip (12V)
- GPIO 13 → Relay 5 / D5 LED: Solenoid Water Valve (12V NC)
- GPIO 14 → Water Flow Sensor Pulse In (YF-S201 Hall)
- GPIO 21 → Capacitive Water Level In (XKC-Y25 Digital)
- GPIO 47 → WiFi Status LED (Blue)
- GPIO 48 → System Alarm LED (Red)
- GPIO 38 → Rotary Mode Selector (Auto/Man)
- GPIO 0 → Setup / Provisioning Button
- EN / RST → Hardware Reset Button

MANDATORY HARDWARE PROTECTION STANDARDS:

1. Inductive Flyback: Fast 1N4007 recovery diodes MUST be placed across all 12V relay coils, fans, pump motor, and solenoid valve.
2. Anti-Corrosion Strobe: Resistive water level sensor power pin (IO8) is energized ONLY during 15ms sampling cycles to prevent electrolysis.
3. Thermal Isolation: MQ-135 5V internal heater coil must maintain at least 10cm separation from DHT11/SHT31 to prevent radiant heat bias.